



TTRL:

Test-Time Reinforcement Learning

Zuo, Yuxin, et al, arXiv 2025

자연어 처리팀

김기환, 조해창, 정현석, 임다예슬, 이웅기, 박민건, 황예진, 박주혜, 이예서, 김유진, 김보성



1 INTRODUCTION

2 METHOD

3 RESULT

4 CONCLUSION

1

INTRODUCTION



1

INTRODUCTION

2

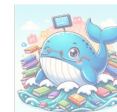
METHOD

3

RESULT

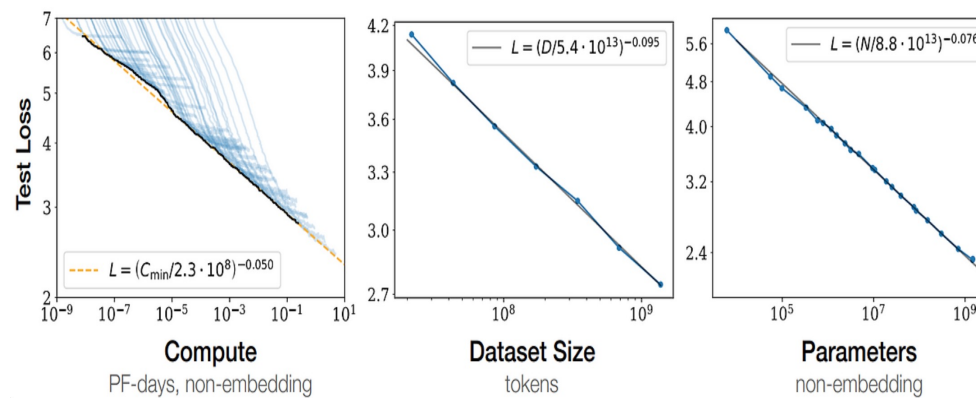
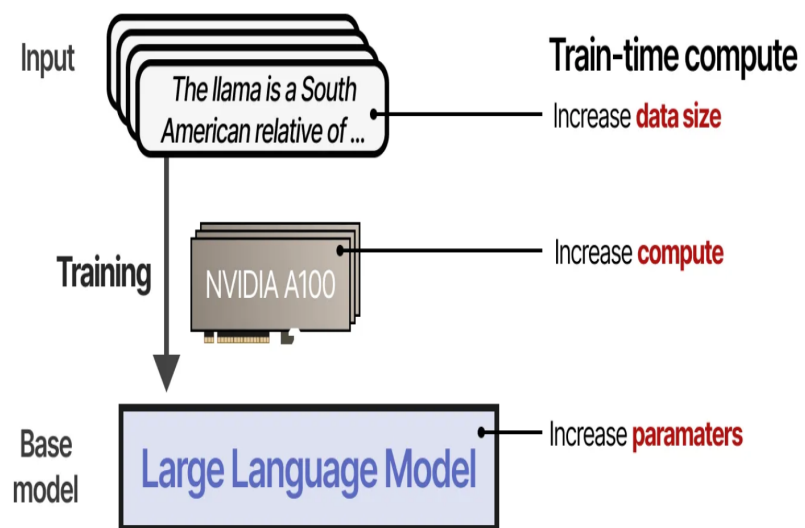
4

CONCLUSION



Train Time Compute

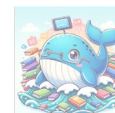
학습 시점에 계산 리소스(Train Time Compute)를 증가시켜 대규모 언어 모델(LLM)의 능력을 향상시키는것



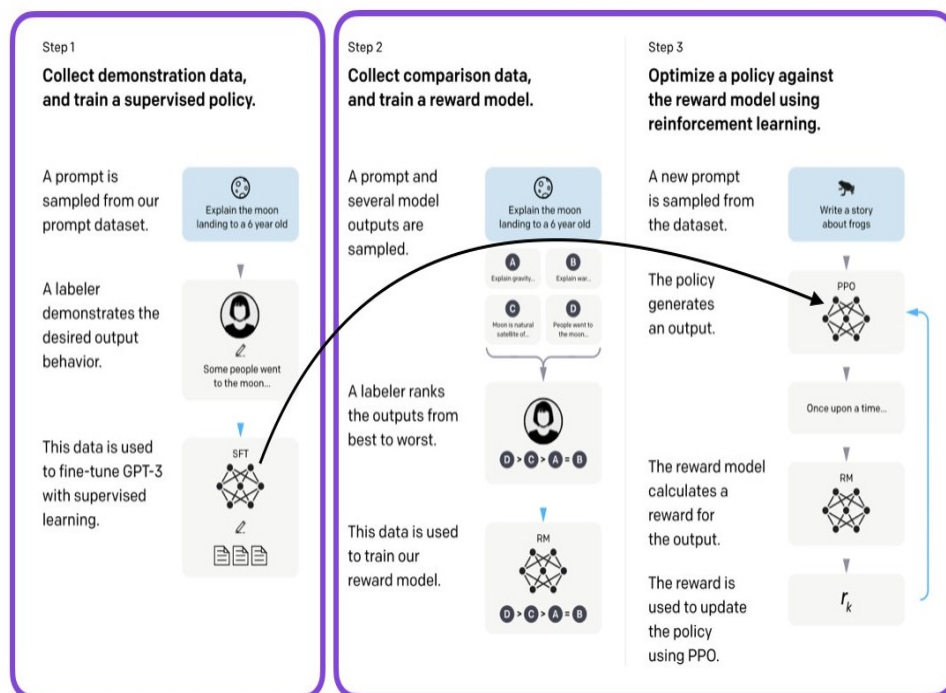
Scaling Laws: Big Model

Chinchilla Scaling Laws: Big Model & Big Data

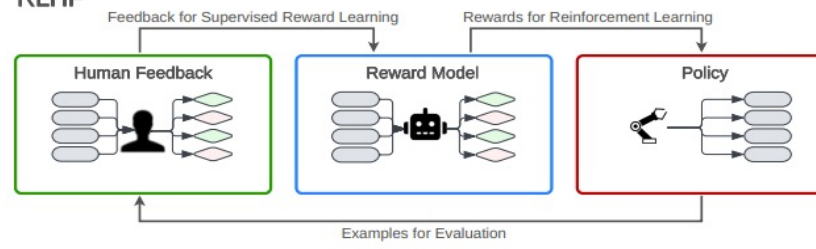




Reinforcement Learning(RL)



RLHF



Challenges

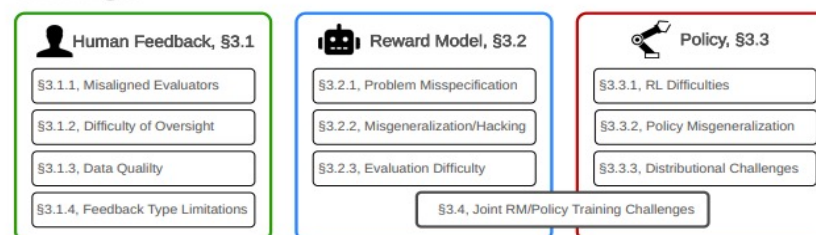
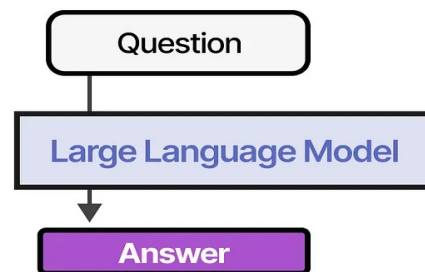


Figure 1: **(Top) Reinforcement Learning from Human Feedback.** Gray, rounded boxes correspond to outputs (e.g., text), and colored diamonds correspond to evaluations. **(Bottom) Our taxonomy for challenges with RLHF.** We divide challenges with RLHF into three main types: challenges with obtaining quality **human feedback**, challenges with learning a good **reward model**, and challenges with **policy** optimization. In the figure, each contains boxes corresponding to the subsections of Section 3.

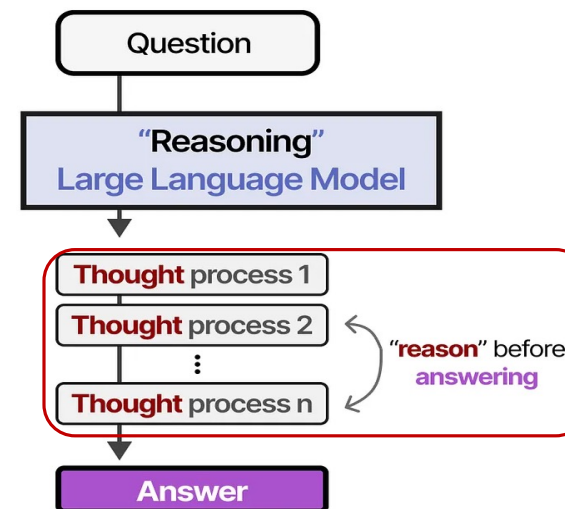


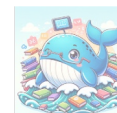
Large Reasoning Models (LRMs)

“Regular” LLMs



“Reasoning” LLMs





Large Reasoning Models (LRMs)

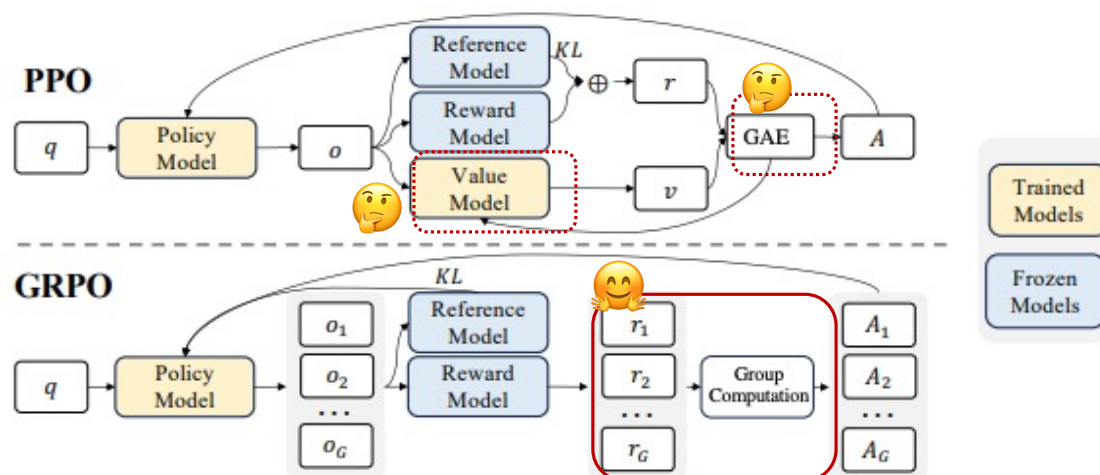
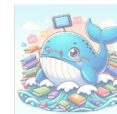
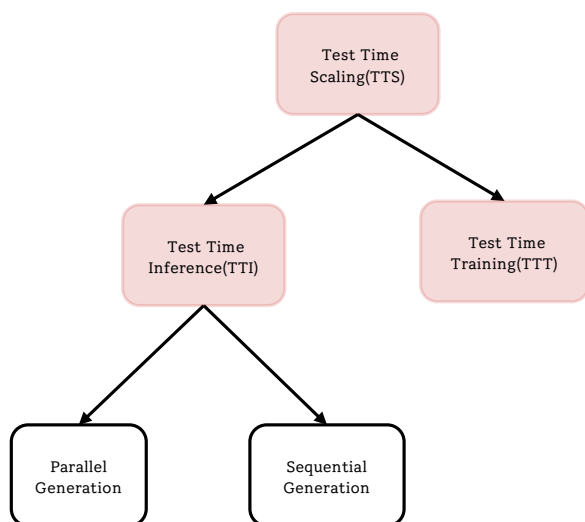


Figure 4 | Demonstration of PPO and our GRPO. GRPO foregoes the value model, instead estimating the baseline from group scores, significantly reducing training resources.



Test Time Scaling

추론 시점에 계산 리소스(**Test-Time Compute**)를 증가시켜 대규모 언어 모델(LLM)의 능력을 향상시키는 것



➤ Test Time Inference

여러 후보 출력을 생성하고 스코어링 함수를 사용하여 최적의 출력을 선택하거나, 반복적인 프로세스를 통해 출력을 개선하는 방식 등을 포함.

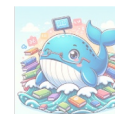
다수결 투표(Majority Voting) 나 Best-of-N과 같은 기법들이 존재

➤ Test Time Training

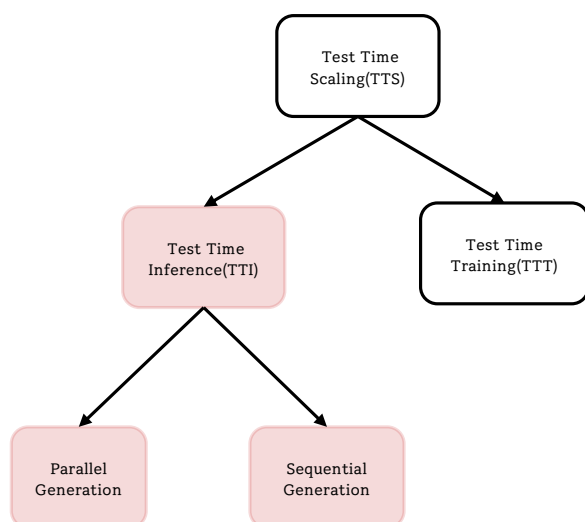
분포 변화(distribution shifts) 하에서 일반화 성능을 향상시키기 위해 추론 시점에 사전 학습된 모델을 적응시키는 기법

테스트 샘플을 사용하여 모델 파라미터를 온라인으로 업데이트하지만, 정답 레이블(ground-truth labels)에 접근하지 않고도 보조적인 자가 지도 학습(self-supervised loss)을 최소화하는 방식으로 학습

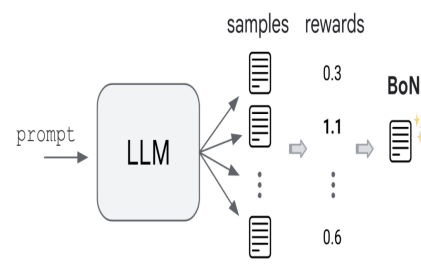
1 INTRODUCTION



Test Time Scaling



➤ Parallel Generation



```
1 def best_of_n(n: int, reward_model):
2     best_reward: float = float('-inf')
3     best_output = None
4     outputs = model(n)
5     for output in outputs:
6         reward = reward_model(output)
7         if reward > best_reward:
8             best_reward = reward
9             best_output = output
10    return best_output, best_reward
11
```

➤ Sequential Generation

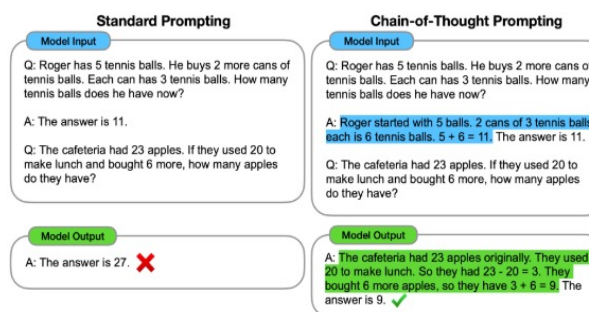
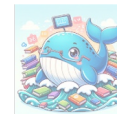
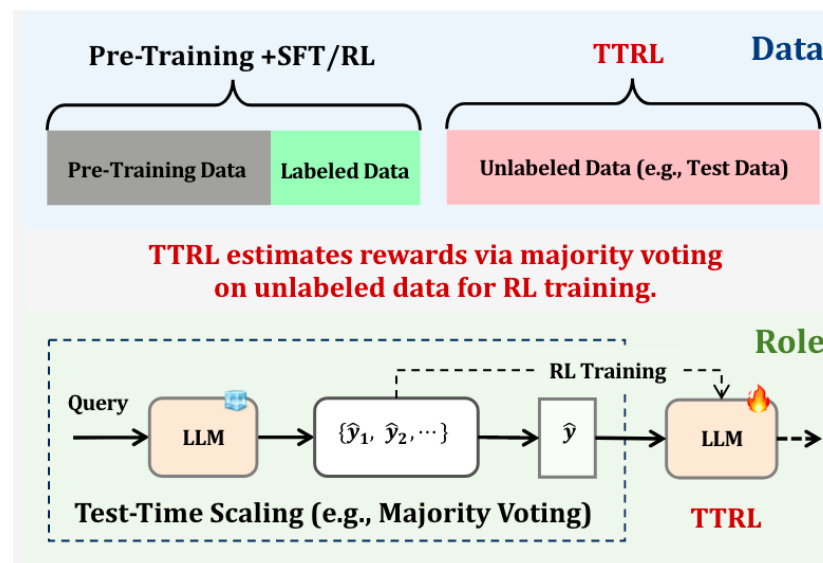
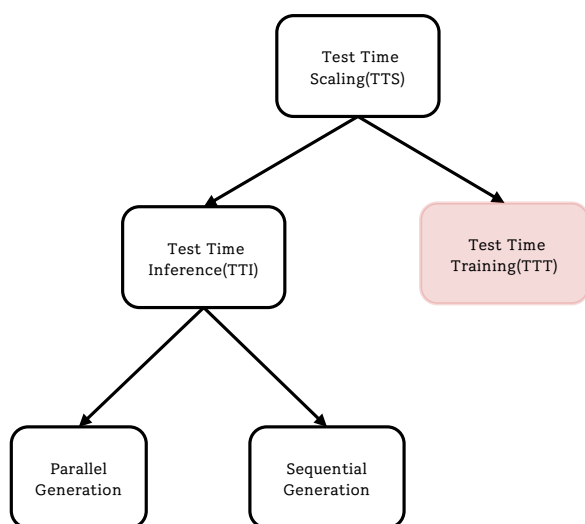


Figure 1: Chain-of-thought prompting enables large language models to tackle complex arithmetic, commonsense, and symbolic reasoning tasks. Chain-of-thought reasoning processes are highlighted.



Test Time Scaling





1 INTRODUCTION

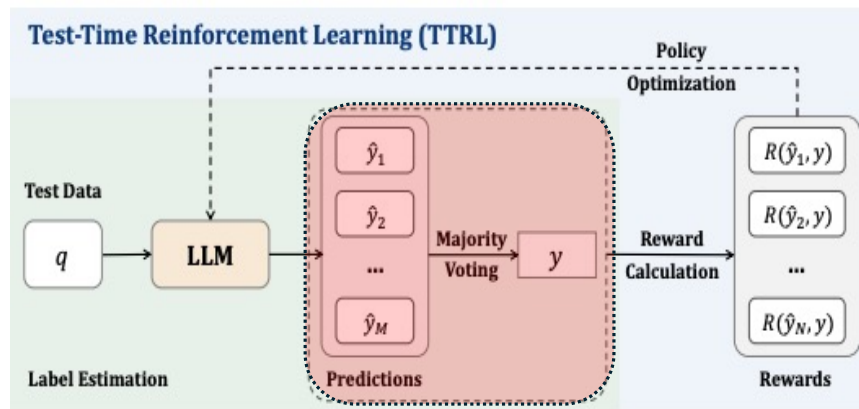
2 METHOD

3 RESULT

4 CONCLUSION



TTRL



1. 주어진 입력(프롬프트)에 대해 모델은 반복적인 샘플링을 통해 여러 개의 후보 출력($\{y_1, y_2, \dots, y_N\}$)을 생성.
2. 이 출력들에 대해 다수결 투표(Majority Voting)와 같은 집계 방법을 사용하여 합의된 출력(consensus output) 또는 추정된 레이블(estimated label) y 를 도출
3. 샘플링된 출력(y)과 추정된 레이블(y^*) 간의 일치도를 기반으로 규칙 기반 보상(rule-based rewards) $r(y, y^*)$ 계산

➤ Test Time Inference

```
from collections import Counter

def majority_voting_reward_fn(outputs):
    """
    Assigns a reward of 1 to each output whose extracted answer matches
    the majority answer, otherwise 0.
    """
    # Extract answers from each output
    answers = [extract_answer(output) for output in outputs]

    # Find the majority answer
    counts = Counter(answers)
    majority_answer, _ = counts.most_common(1)[0]

    # Assign rewards: 1 if matches majority, else 0
    rewards = [1 if ans == majority_answer else 0 for ans in answers]
    return rewards

outputs = llm.generate(problem, n=N)
rewards = majority_voting_reward_fn(outputs)
```

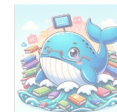
- Consensus = majority voting = ground truth

➤ Test Time Training

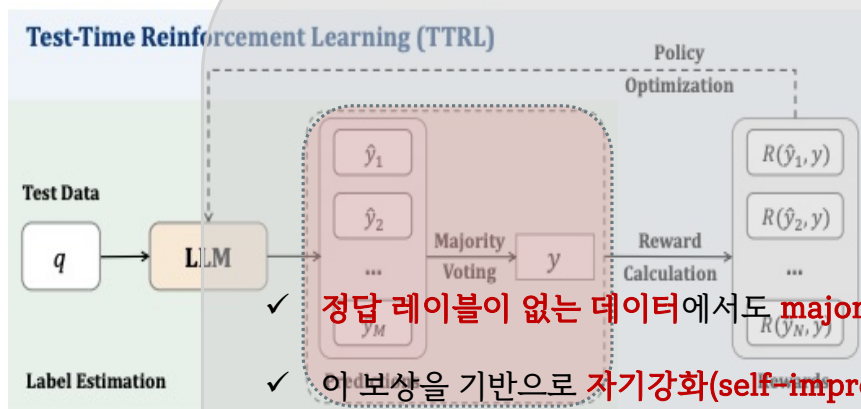
$$\max_{\theta} \mathbb{E}_{y \sim \pi_{\theta}(\cdot|x)} [r(y, y^*)]$$

$$\theta \leftarrow \theta + \eta \nabla_{\theta} \mathbb{E}_{y \sim \pi_{\theta}(\cdot|x)} [r(y, y^*)],$$

- Gradient Ascent



TTRL



✓ **정답 레이블이 없는 데이터에서도 majority voting을 통해 보상을 추정하고,**

✓ **이 보상을 기반으로 자기강화(self-improvement) 학습을 수행하는 것**

✓ **즉, 레이블 없이도 보상을 통해 강화학습을 수행할 수 있도록 설계**

1. 주어진 입력(프롬프트)에 대해 모델은 반복적인 샘플링을 통해 여러 개의 후보 출력($\{y_1, y_2, \dots, y_N\}$)을 생성.
2. 이 출력들에 대해 다수결 투표(Majority Voting)와 같은 집계 방법을 사용하여 합의된 출력(consensus output) 또는 추정된 레이블(estimated label) y 를 도출
3. 샘플링된 출력(y)과 추정된 레이블(y^*) 간의 일치도를 기반으로 규칙 기반 보상(rule-based rewards) $r(y, y^*)$ 계산

➤ Test Time Inference

```
from collections import Counter

def majority_voting_reward_fn(outputs):
    """
    Assigns a reward of 1 to each output whose extracted answer matches
    the majority answer, otherwise 0.
    """
    # Extract answers from each output
    answers = [extract_answer(output) for output in outputs]

    # Find the majority answer
    counts = Counter(answers)
    majority_answer = counts.most_common(1)[0][0]

    # Assign rewards: 1 if matches majority, else 0
    rewards = [1 if ans == majority_answer else 0 for ans in answers]
    return rewards

outputs = llm.generate(problem, n=N)
rewards = majority_voting_reward_fn(outputs)
```

- Consensus = majority voting = ground truth

➤ Test Time Training

$$\max_{\theta} \mathbb{E}_{y \sim \pi_{\theta}(\cdot|x)} [r(y, y^*)]$$

$$\theta \leftarrow \theta + \eta \nabla_{\theta} \mathbb{E}_{y \sim \pi_{\theta}(\cdot|x)} [r(y, y^*)]$$

- Gradient Ascent

3

RESULT



1

INTRODUCTION

2

METHOD

3

RESULT

4

CONCLUSION



Main Result

Table 1: Main results of TTRL on each task. * indicates results from Dr. GRPO (Liu et al., 2025b). Our training data size matches the corresponding benchmark dataset size.

Name	AIME 2024	AMC	MATH-500	Avg	Labeled Data
Qwen2.5-Math-1.5B*	20.0	32.5	33.0	28.5	-
w/ TTRL	20.0	53.0	80.0	51.0	✗
Δ	0	+20.5	+47.0	+22.5	✗
	0	$\uparrow 63.1\%$	$\uparrow 142.4\%$	$\uparrow 79.0\%$	✗
Qwen2.5-Math-1.5B-Instruct	10.0	48.2	74.2	44.1	3.1M
DeepSeek-R1-Distill-1.5B@3k	2.5	21.7	52.2	25.5	800K
DeepSeek-R1-Distill-1.5B@8k	20.0	49.4	77.4	48.9	800K
Oat-Zero-1.5B	20.0	53.0	74.2	49.1	8.9K
Qwen2.5-Math-7B*	16.7	38.6	50.6	35.3	-
w/ TTRL	43.3	67.5	84.2	65.0	✗
Δ	+26.6	+28.9	+33.6	+29.7	✗
	$\uparrow 159.3\%$	$\uparrow 74.9\%$	$\uparrow 66.4\%$	$\uparrow 84.1\%$	✗
Qwen2.5-Math-7B-Instruct*	16.7	53.0	83.6	51.1	3.1M
DeepSeek-R1-Distill-7B@3k*	10.0	26.2	60.1	32.1	800K
SimpleRL-Zero-7B*	26.7	60.2	78.2	55.0	8.9K
PRIME-Zero-7B*	16.7	62.7	83.8	54.4	230K
OpenReasoner-Zero-7B@3k*	13.3	47.0	79.2	46.5	129K
Oat-Zero-7B*	43.3	62.7	80.0	62.0	8.9K
LIMR-7B	32.5	63.8	78.0	58.1	1.4K
LLaMA-3.1-8B-Instruct	3.3	19.3	47.8	23.5	-
w/ TTRL	3.3	32.5	61.8	32.5	✗
Δ	0	+13.2	+14.0	+9.0	✗
	0	$\uparrow 68.4\%$	$\uparrow 29.3\%$	$\uparrow 38.5\%$	✗

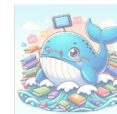
and adopt the AdamW optimizer for the policy model. For rollout, we sample 64 responses (32 for MATH-500) using a temperature of 1.0 for voting-based label estimation and down-sample 16 responses per prompt for training. Evidence shows that our vote-then-sample strategy effectively reduces computational costs while still achieving strong performance. The maximum number of tokens for generation is set to 3072. The KL coefficient is set to 0 across all experiments. We set the number of episodes to 40, 50, and 60 for MATH, AMC, and AIME, respectively, based on the dataset size and task complexity.

➤ TTRL performs well on most tasks and models

Despite relying solely on self-evolution using unlabeled test data, TTRL achieves performance comparable to existing RL-based models trained on large-scale labeled datasets. As shown in Table 1, on the highly challenging mathematical reasoning benchmark AIME 2024, TTRL achieves a substantial improvement of 159.3%, surpassing all models trained on large-scale datasets. Furthermore, when applied to Qwen2.5-Math-7B, TTRL yields an average improvement of 84.1% across three benchmarks.

➤ TTRL naturally scales

Another noteworthy observation is that as the model size increases (from 1.5B to 7B), performance gains on both AIME 2024 and AMC increase, highlighting the natural scaling behavior of TTRL: larger models can produce more accurate majority voting rewards during self-improvement, which leads to more effective learning on new data. However, LLaMA-3.1-8B-Instruct and Qwen2.5-Math-1.5B fail to achieve meaningful gains on AIME 2024 through TTRL, likely due to their limited capacity. In contrast, Qwen2.5-Math-7B's greater model capacity and more sufficient knowledge allows it to benefit from self-improvement, resulting in clear performance gains. We discuss this in more detail in Section 4.3.



Main Result

➤ TTRL generalizes well beyond the target task (Out of distribution)

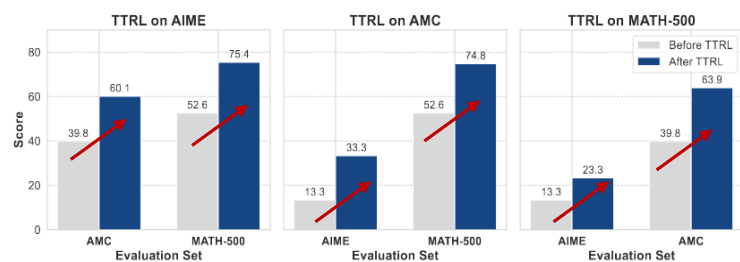


Figure 3: Out-of-distribution performance before and after TTRL.

We perform TTRL on each benchmark and further evaluate on others, with Qwen2.5-Math-7B as the backbone. Figure 3 shows the results. Despite the out-of-distribution nature of this setting, TTRL achieves substantial improvements across all benchmarks. This suggests that TTRL does not rely on overfitting, which would lead to trade-offs on other tasks, but instead acquires generalizable gains during self-improvement.

➤ TTRL is compatible with different RL Algorithms

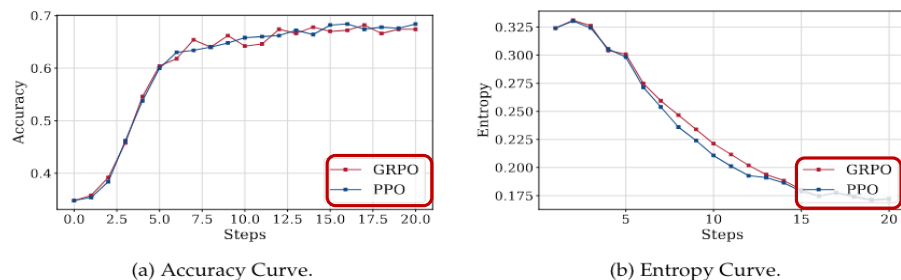
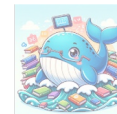


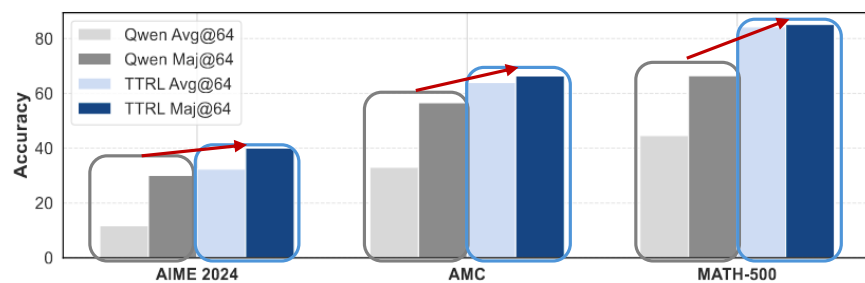
Figure 4 presents the results. We apply TTRL using PPO (Schulman et al., 2017) on MATH-500 to assess its compatibility with different reinforcement learning algorithms. The performance trajectories of PPO and GRPO are closely aligned. Compared to GRPO, PPO yields more stable outcomes while achieving similar overall performance.

Figure 4: Comparison over steps of different RL algorithms, GRPO vs PPO on MATH-500.



Discussion

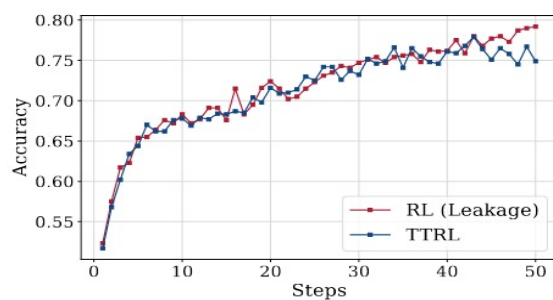
➤ Q1: How well can TTRL perform?



- TTRL is Supervised by Maj@N Yet Surpasses It.

1. TTRL은 backbone 모델(Qwen)보다 훨씬 더 높은 정확도를 달성한다
2. 평균 정확도(Avg@64)보다 Majority Voting(Maj@64)를 쓰면 성능이 더 좋아진다
3. 세 가지 벤치마크(AIME, AMC, MATH-500) 모두에서 이 경향이 일관되게 나타난다

Figure 6: Majority voting performance comparison between backbone and TTRL model.



- ground-truth label로 직접 RL을 수행한 모델과 정답 정보를 일부러 노틀한 RL(Leakage)과 비교시 근접

Figure 7: Comparison of RL (Leakage) vs TTRL.



Discussion

➤ Q2: Why Does TTRL Work?

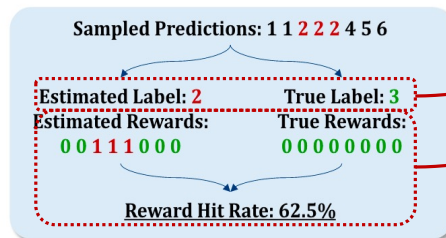


Figure 9: A toy case. We illustrate a basic numerical prediction scenario to compare reward computation under two conditions: when the model incorrectly estimates the label versus when the ground-truth label is used. As shown on the left, although the estimated label is incorrect, some of the incorrect predictions still differ from the wrong label and therefore receive the correct reward (denoted as 0).

“잘못된 정답(incorrect label estimate)”을 기반으로 학습하는데도 왜 여전히 효과적인가

- Reward 는 정확한 정답이 아닌 **일관된 기준에 의존**
- Reward는 label(ground-truth) 보다 dense feedback 진행.
- Base 성능이 안 좋더라도 오히려 성능이 올라가는 현상 발생,

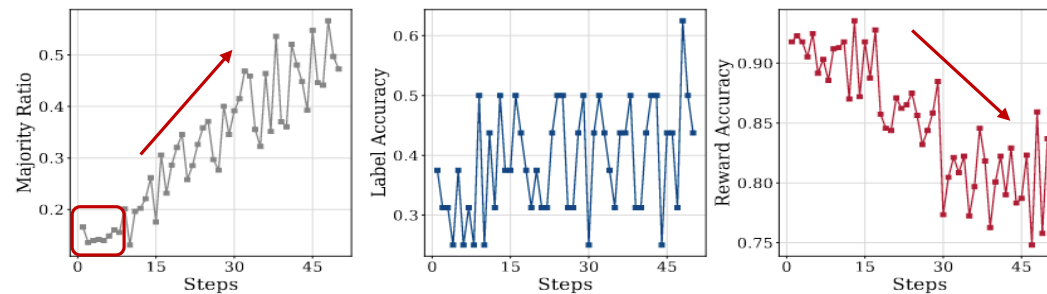
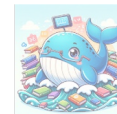


Figure 8: Comparison of Majority Ratio, Label Accuracy, and Reward Accuracy.

- **Majority Ratio:** 하나의 rollout 내에서 가장 자주 등장한 응답의 비율
- **Label Accuracy(Maj@N):** 추정한 label (majority voting)이 실제 정답과 일치 비율
- **Reward Accuracy:** 추정한 label로 계산한 reward와 ground-truth label로 계산한 reward가 일치하는 비율



Discussion

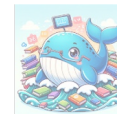
➤ Q3: When Might TTRL Fail?

➤ Lack of Prior Knowledge on Target Task

Table 2: Performance of **TTRL** across the five difficulty levels of MATH-500.

Metric	Name	MATH-500-L1	MATH-500-L2	MATH-500-L3	MATH-500-L4	MATH-500-L5
Accuracy	Backbone	25.9	33.0	36.3	32.5	22.3
	w/ TTRL	71.2	76.2	76.3	58.7	39.2
	Δ	+45.4	+43.2	+40.0	+26.2	+16.8
		↑175.3%	↑130.8%	↑110.2%	↑80.4%	↑75.3%
Response Len.	Backbone	2,339.2	2,125.1	2,120.6	1,775.1	1,751.3
	w/ TTRL	624.3	614.4	672.3	783.5	985.3
	Δ	-1,715.0	-1,510.6	-1,448.3	-991.6	-766.0
		↓73.3%	↓71.1%	↓68.3%	↓55.9%	↓43.7%

- 사전 지식 부족
- 데이터 난이도에 민감
- 특정 조건 하에서 collapse 가능성



Discussion

➤ Q3: When Might TTRL Fail?

➤ Inappropriate RL Hyperparameters

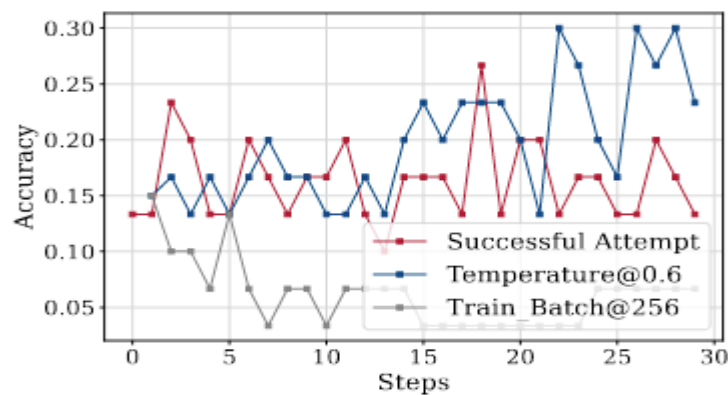


Figure 10: Failed attempts. We compare the curves under settings with appropriate parameters versus those with suboptimal temperature and training batch size. The results show that the performance is particularly sensitive to the training batch size in such settings.

RL 자체가 학습 환경에 민감

test-only, noisy reward 환경이기 때문에 더 중요.

- Temperature: 어려운 태스크일 수록 ▲
- Episodes: 데이터 셋이 작거나 어려울수록 ▲



1 INTRODUCTION

2 METHOD

3 RESULT

4 CONCLUSION



- TTRL은 정답이 없는 테스트 데이터만으로도 모델의 자기 개선(self-improvement)을 가능하게 하는 새로운 강화학습 패러다임 제안.
- AIME 2024와 같은 고난도 벤치마크에서 Qwen2.5-Math-1.5B 및 LLaMA-3.1-8B-Instruct와 같이 사전 지식이 부족한 소형 모델에게 효과가 제한적.
- TTRL은 RL 학습의 특성상 하이퍼파라미터 설정에 민감하여 temperature와 episodes와 같은 하이퍼파라미터는 학습 안정성과 성공에 결정적인 영향.
- 향후에는 다른 다양한 태스크에서도 적용 가능한지 실험 필요
- Online Learning with Streaming Data(Test-Time Adaptation)
- Large-Scale Self-Supervised RL Training



감사합니다.

자연어 처리팀

김기환, 조해창, 정현석, 임다예슬, 이웅기, 박민건, 황예진, 박주혜, 이예서, 김유진, 김보성